Predicting AI Results on New Benchmarks with a Gaussian Process IRT Model Fit at Test Time

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Abstract

Running every AI system on every benchmark costs a lot, so it helps to predict how a system will do on a new benchmark from only a few of its results. The Predictive AI Evaluation Competition (PAIEC) asks for a predictor that gives the chance that a system answers an item correctly, when the benchmark is not in the training data and labels are shown a few at a time (0, 1, 3, 7, 15 and 31 per system and benchmark). This is hard because, with so few labels, most of the error comes from not knowing how well the system does on the benchmark as a whole. To address this challenge, we design GP-IRT, which fits a logistic item response model on each test benchmark at test time, using the labels of all systems on that benchmark. Item effects have a Gaussian process prior over item embeddings, so items with similar text get similar difficulty, and predictions are averaged over a grid of prior scales, with weights from the Laplace marginal likelihood. The prior on the benchmark level is centred on the accuracy of groups of the size used at test time, which is about 0.36 in training and 0.35 in the test groups we recovered from formative feedback. We test GP-IRT with a simulator of the official protocol on held-out benchmarks. GP-IRT reaches a Brier ALC of 0.1797, against 0.1869 for a group mean and 0.25 for a constant 0.5. Centring the prior gives the largest single gain: it lowers the Brier score at budget 0 from 0.248 to 0.236 and the Brier ALC by 0.002.

1 Introduction

Benchmarks are the main way to compare AI systems, but running a new system on many benchmarks takes time and money. Predictive evaluation tries to guess the result of a system on an item before running it, from results we already have [AIMS Lab, Stanford University, 2026]. PAIEC asks this question in a hard setting: the test benchmarks never appear in training, and the predictor sees only a few labels on each of them. The training data is the AI Measurement Data Bank [Truong et al., 2026], and predictions are scored with the Brier score [Brier, 1950], averaged over label budgets.

Limitation of existing solutions: Item response theory (IRT) [Lord, 2012] gives the chance of a correct answer from the ability of a subject and the difficulty of an item, and it has been used to evaluate language models on a few items. For example, tinyBenchmarks [Maia Polo et al., 2024] estimates the accuracy of a new model from a small set of items, with item parameters learned from many models already run on the same benchmark. In PAIEC the test benchmarks are new, so the item parameters are not known in advance and must be learned at test time from the few labels shown. Simple baselines, such as the mean of the labels shown for a group, do not use the labels of other systems or the text of the items.

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Challenges: New benchmarks with very few labels raise four problems: ① *Unknown benchmark level:* with zero or one label, most of the error comes from the overall level of the system on the benchmark, not from single items. ② *Few labels per item:* most items have no label, so their difficulty must be learned from the labels of other systems and from the item text. ③ *Benchmarks differ a lot:* on some benchmarks a system gets almost every item right or almost every item wrong, while on others items differ little, so one prior does not fit all of them. ④ *No test benchmark to choose a model:* the formative feedback on Codabench [Xu et al., 2022] is computed on a new random draw for each submission, so it cannot be used to compare models.

Contribution: In this paper, we make the following contributions:

- To address challenge ①, GP-IRT puts a prior on the benchmark shift that is centred on the level of training groups of realistic size. The formative logs show that the test groups have the same low accuracy (0.35 on average).
- To address challenge ②, GP-IRT uses the labels of all systems on a benchmark together and gives the item effects a Gaussian process prior [Rasmussen and Williams, 2006]. Its kernel combines the similarity of text embeddings and of metadata tokens with a linear kernel on embeddings centred within the benchmark.
- To address challenge ③, GP-IRT fits the model for a grid of prior scales and averages the predictions, with weights from the Laplace marginal likelihood, as in Bayesian model averaging [Hoeting et al., 1999].
- To address challenge ④, we built a simulator of the official protocol, with 5-fold cross-validation over benchmarks and paired comparisons, and made every design choice with it. We release the code and the simulator.

2 Task and Data

Task: A subject is an AI system with fixed settings. An item is one question, with its text, some metadata and an anonymous benchmark ID. A group is one subject on one benchmark. In each scored group with at least 80 items, the items are split in two halves: labels are shown from one half, and the other half is scored. Labels come in nested steps of 0, 1, 3, 7, 15 and 31 per group. The predictor sees the labels of every group, including groups it does not predict. The score is the Brier ALC, $0.1B_0 + 0.2B_1 + 0.2B_3 + 0.2B_7 + 0.2B_{15} + 0.1B_{31}$, averaged over groups.

Training data: We use only the public training pool `aims-foundations/measurement-db` (revision `54074453`). We keep benchmarks with binary item-level answers and `release = public`, which gives 160 benchmarks. Of these, 92 have at least one group with 80 to 1000 items, for a total of 10,208 groups, 154,628 items and 4.3M subject and item cells. Each cell holds n trials and k successes. No other evaluation data is used.

3 GP-IRT: Item Response Prediction Fit at Test Time

3.1 Overview

For each test benchmark, GP-IRT works in five steps. It first collects all labels shown on the benchmark, from any subject (Step 1). It then builds item vectors from the item text and metadata (Step 2). It fits a logistic item response model with a Gaussian process prior on the item effects, once for each point of a grid of prior scales (Step 3). It averages the predictions of the grid, with weights from the Laplace marginal likelihood (Step 4). Finally, it returns the chance that the subject answers each target item correctly (Step 5). The fits are redone each time new labels arrive.

3.2 Model

Fix a benchmark b and let $\mathcal{L}$ be the labels shown on b, from any subject. For subject s and item t,

$$\operatorname{logit} P(y_{st} = 1) = c_t + m + a_s + d_t, \quad a_s \sim \mathcal{N}(0, \sigma_a^2),\ d \sim \mathcal{GP}(0, \sigma_d^2 K). \tag{1}$$

Here c_t is a fixed offset from a prior model (Section 3.6), m is a shift for the benchmark, a_s is a shift for the subject, and d_t is the item effect, shared by all subjects of the benchmark. The prior on m is

given in Section 3.5. The item kernel is

$$K = (1-\rho_\ell)\big[\rho\, e^{(\cos(e_i,e_j)-1)/\ell} + \rho_f\, \kappa_{\text{tok}}(i,j) + (1-\rho-\rho_f)\mathbb{1}[i=j]\big] + \rho_\ell \frac{\tilde{e}_i^\top \tilde{e}_j}{\|\tilde{e}\|^2}. \qquad (2)$$

e is the normalised `multilingual-e5-small` [Wang et al., 2024] embedding of the item text and metadata. κ_{tok} is the cosine similarity of `key=value` metadata tokens, counting only tokens found on at least two labeled items. $\tilde{e} = e - \bar{e}_{\mathcal{L}}$ is the embedding minus its mean over the labeled items of the benchmark. The last term acts as a Bayesian linear regression of item difficulty on the embedding, learned for each benchmark from its labels.

3.3 Inference

We find the posterior mode with Newton's method. The Hessian block of the subject effects is diagonal, so we remove it with a Schur complement. What is left is a block for m and the labeled item effects, of size at most the number of distinct labeled items. One Newton step costs $O(|\mathcal{L}| + n_s n_{\mathcal{L}} + n_{\mathcal{L}}^3)$. The Laplace posterior [Rasmussen and Williams, 2006, Chapter 3] gives a mean μ and a variance v of the logit for any target item, labeled or not, through the Gaussian process conditional of d_t given $d_{\mathcal{L}}$. We predict $\sigma\big(\mu/\sqrt{1+\pi\gamma v/8}\big)$. With $\gamma = 1$ this is the usual probit approximation of the predictive integral [MacKay, 1992]. We use $\gamma = 0.35$ while the target subject has at most three labels on the benchmark and $\gamma = 0.2$ after that (Section 5).

3.4 Averaging over Prior Scales

We fit the model for every combination of $\sigma_a \in \{0.4, 0.8, 1.6\}$, $\sigma_d \in \{0.8, 1.6, 3.2\}$ and the three parts of the shift prior below, with $\rho = 0.73$, $\rho_f = 0.17$, $\ell = 0.12$ and $\rho_\ell = 0.4$. The predicted chances are averaged with weights proportional to the Laplace marginal likelihood. This is done for each benchmark and each set of labels separately.

3.5 Prior on the Benchmark Shift

The shift m sets the overall level of a subject on a new benchmark. Its prior mixes three normal distributions with equal weights,

$$m \sim \tfrac{1}{3}\big[\mathcal{N}(\mu_0, 0.8^2) + \mathcal{N}(\mu_0, 1.6^2) + \mathcal{N}(\mu_0, 3.2^2)\big], \qquad \mu_0 = -0.6, \qquad (3)$$

and each part is one more axis of the averaging grid. The wide part lets the model follow benchmarks where a subject gets almost everything right or wrong, and the narrow part keeps predictions stable with one or two labels. The centre μ_0 matters more than the scales. In training groups of realistic size, the mean logit accuracy is -0.87, with a standard deviation of 1.8, so a prior centred at zero starts every new benchmark from an accuracy that is too high. We chose μ_0 by cross-validation (Section 5). The chosen value lies between zero and the training mean, because the offset c_t, whose mean is negative, already accounts for part of the low level, and the probit correction pulls predictions toward 0.5.

3.6 Prior across Benchmarks and Second Model

We take hidden states of Qwen2.5-7B-Instruct [Qwen Team et al., 2025] (layer 21 of 28, mean over tokens, standardised) and fit a ridge regression to the mean logit accuracy of each item, giving the same weight to every training benchmark. The output is z_t, and we set $c_t = 0.3\, z_t$. With no labels, the prediction is the prior predictive, averaged over the parts of the shift prior. The final predictor is the average of two copies of the model that differ only in the item vectors used by the kernel: e5-small embeddings in one, and Qwen hidden states in the other.

3.7 Implementation

Labels are taken in the platform's default random order (Section 6). Without a GPU, or if Qwen cannot be loaded or fails during the run, the predictor uses the e5 model alone, with no offset c_t. In simulation, this costs only 0.0003 ALC in the realistic setting. A prediction takes 15 to 110 ms on one A100 GPU.

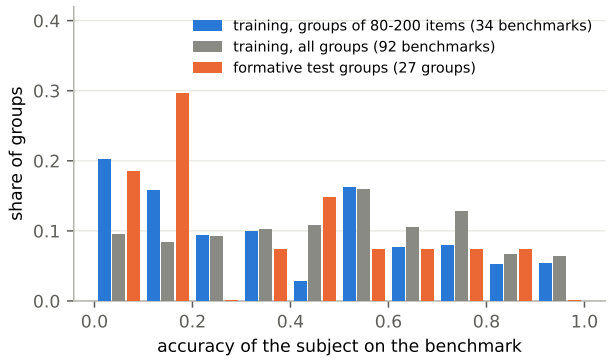


Figure 1: Accuracy of a subject on a benchmark, for training groups and for test groups seen in formative feedback.

4 Experimental Setup

We built GP-IRT in Python. The Qwen hidden states took about 2.5 hours on one A100 80GB GPU, and the simulations ran on 30 CPU cores, where 1000 episodes of the realistic setting take 10 to 20 minutes.

Simulator: Test benchmarks never appear in training, so we made every choice with a simulator of the official protocol and 5-fold cross-validation over benchmarks. Groups are added to a draw one at a time. Groups whose benchmark or model name is already in the draw come last, and ties are broken at random. We stop when the draw would pass 1000 subject and item pairs. In each group, the distinct items are split at random, and the odd item goes to the label pool. Labels come in random order, with one sampled trial per item, and every group sees the labels of all other groups.

Settings: We use three settings. In the realistic setting, our main one, only groups with at most 200 items are allowed. A draw then has on average 6.9 groups, 5.9 benchmarks and 142 items per group. The large group setting allows all groups with 80 to 1000 items and gives about 3 groups per draw. In the many subjects setting, the limit counts distinct items, so a benchmark can have hundreds of subjects.

Baselines: We compare GP-IRT with a constant 0.5, the mean of the labels shown for the group with a Beta(1,1) prior, and the Bayes predictor of the group accuracy from the group's own labels, with the accuracies of training groups from other folds as its prior (groups of 80 to 200 items, in all three columns). We also compare with IRT models without item effects and with independent item effects.

Metrics: We compute the exact Brier score of each group over all its response rows, apply the ALC weights, and average over groups and then over episodes. Two models are always compared on the same episodes, with the same splits and the same labels, and we report the paired mean difference with its standard error. Settings were chosen on one seed and checked again on episodes drawn with another seed.

Formative feedback: Each Codabench submission is scored on a new draw of eight or nine groups, and the scoring log gives the Brier score and the calibration error (ECE) of each group at each budget. Since the draw changes, these logs say little about which model is better, but they do describe the test groups. A draw has about nine groups from seven benchmarks, with 88 to 152 items per group, which matches the realistic setting. At budget 0 the prediction is the same for all items of a group, so the ECE is the distance between this prediction and the group accuracy, and the Brier score tells on which side of the prediction the accuracy lies. In this way we recovered the accuracy of 27 test groups, with a mean of 0.35. Fig. 1 compares them with the training groups. The test groups look like the training groups of realistic size (mean 0.36 per group, 0.39 per benchmark), and not like the training pool as a whole (0.49 per benchmark), where large benchmarks with many subjects count the most. The logs led us to the realistic setting and to the level of the shift prior. Every parameter, including the centre of that prior, was then chosen by cross-validation on the training pool.

Table 1: Simulated Brier scores with held-out benchmarks (1000 episodes per setting, the same in every row).

Model	realistic B_0	B_1	B_3	B_7	B_{15}	B_{31}	ALC	large ALC	many ALC
Constant 0.5	.2500	.2500	.2500	.2500	.2500	.2500	.2500	.2500	.2500
Group mean, Beta(1,1)	.2500	.2121	.1862	.1708	.1619	.1572	.1869	.2083	.2133
Bayes, own labels, empirical prior	.2356	.2072	.1856	.1715	.1626	.1576	.1847	.2089	.2129
IRT, no item effects	.2500	.2077	.1835	.1698	.1616	.1571	.1852	.2096	.2047
IRT, independent item effects	.2500	.2113	.1883	.1741	.1652	.1599	.1888	.2091	.1799
GP-IRT, e5 RBF kernel	.2500	.2083	.1846	.1700	.1605	.1540	.1851	.2075	.1738
+ metadata token kernel	.2500	.2090	.1839	.1684	.1583	.1510	.1840	.2071	.1728
+ linear kernel, centred emb.	.2500	.2091	.1856	.1688	.1582	.1507	.1844	.2046	.1726
+ Qwen offset c_t	.2483	.2082	.1851	.1685	.1580	.1507	.1839	.2041	.1729
+ second model, Qwen kernel	.2483	.2083	.1852	.1684	.1578	.1502	.1838	.2039	.1709
+ tuned prior scales and γ	.2483	.2055	.1829	.1672	.1571	.1497	.1824	.2038	.1701
+ shift scale mixture, γ by labels	.2483	.2046	.1822	.1668	.1568	.1494	.1819	.2046	.1698
+ shift prior centred at -0.6 (final)	**.2356**	**.2012**	**.1814**	**.1666**	**.1567**	**.1494**	**.1797**	.2041	**.1693**

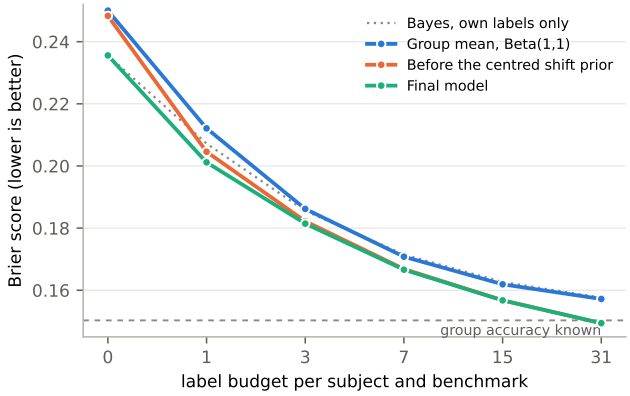


Figure 2: Brier score by budget in the realistic setting.

5 Evaluation Results

We answer four research questions (RQs): (1) how well GP-IRT predicts results on new benchmarks compared with the baselines; (2) what each part of the model adds; (3) how the prior on the benchmark shift changes the results; and (4) how risky predictions are with very few labels.

RQ1: How well does GP-IRT predict results on new benchmarks compared with the baselines?: Table 1 gives the Brier scores of GP-IRT and the baselines. GP-IRT has the lowest ALC in the realistic and many subjects settings (0.1797 and 0.1693), against 0.1869 and 0.2133 for the group mean. With large groups, it is within 0.0003 of the best row, which is inside the noise of that setting. The Bayes reference is useful here. At budget 0 it predicts the mean training accuracy, which is the best constant for the training data. Its B_0 is 0.2356, and GP-IRT reaches the same value, so there is little left to gain at budget 0 without information on single items. From budget 1 on, GP-IRT beats this reference by 0.004 to 0.008 at every budget. This gain comes from the labels of other subjects on the same benchmark and from the item kernel. Fig. 2 shows the curves. Two oracles give limits in the realistic setting. If the accuracy of each group were known, a constant prediction per group would score 0.1503. If the accuracy of each item for each subject were known, the Brier score would be 0.011. Our B_{31} of 0.1494 is just below the first oracle: with 31 labels, the model knows the group level well and gets a little more from the items, but the gap to the second oracle shows how much item information it still misses. On Codabench, the final model (submission 1018959) scored 0.182 Brier ALC, with an ECE of 0.139. Each submission is scored on a new draw of eight or nine groups, so this score can move by several hundredths from one draw to the next.

RQ2: What does each part of the model add?: Table 1 adds the parts of the model one at a time, on the same 1000 episodes in every row. The paired standard error of the difference between two rows is about 10^{-4} in the realistic setting. The centred shift prior gives the largest step in the realistic setting (-0.0022, from 0.1819 to 0.1797), almost all of it at budgets 0 and 1. The centred linear kernel gives the largest step with large groups (-0.0025) and helps from budget 7 on. Most other rows change the ALC by less than 0.001.

RQ3: How does the prior on the benchmark shift change the results?: We compared centres and scales on 300 paired episodes of the realistic setting, against the model of the row before the final one in Table 1. Centres of -0.4, -0.6 and -0.8 lower the ALC by 0.0018 to 0.0022, all with a standard error near 0.0002, while -1.2 gives only 0.0010. The scales matter less: $(1, 2, 4)$, $(0.8, 1.6, 3.2)$ and $(1.5, 3)$ are within 0.0001 of each other. Gaussian mixtures with two or three parts fit directly to the training group accuracies did about as well (-0.0023), so we kept the simpler prior. Once the prior is centred, the Qwen offset adds almost nothing: setting $c_t = 0$ changes the ALC by $+0.00001 \pm 0.00007$. On 1000 new episodes drawn with another seed, the final model beats the row before it by 0.0022 ± 0.0001 ALC, and its Brier score is lower by 0.012 at budget 0 and by 0.004 at budget 1. In the large group setting, where the training accuracy is higher, the centred prior neither helps nor hurts: the change is -0.0008 ± 0.0005 on the episodes used for the choice and $+0.0002 \pm 0.0003$ on new episodes.

RQ4: How risky are predictions with very few labels?: With one to three labels, a predictor can move away from the truth. In simulation, $B_3 > B_0$ happens for about a quarter of the groups, and in 53 groups of formative logs it happened for 24.5%, so the simulator matches the real behaviour on this point. A smaller variance scale γ makes predictions more confident. It lowers the mean ALC down to about $\gamma = 0.1$, but it also makes bad groups more common: the share of groups with $B_3 > 0.3$ grows from 5.2% at $\gamma = 0.5$ to 7.4% at $\gamma = 0.2$. We use $\gamma = 0.35$ while the subject has at most three labels, and $\gamma = 0.2$ after that, when the posterior knows more.

6 Discussion

While GP-IRT does better than the baselines in our simulation, it has some limits:

Simulator: The results depend on how we read the draw rule of the competition. We matched the simulator to the formative logs, and for this reason we checked every choice in three settings.

Level of the test groups: The centred shift prior helps because test groups have a low accuracy on average, like training groups of the same size. On a test set made mostly of easy benchmarks, it would lose part of its gain at budget 0, although it did not hurt in our large group setting.

Model: The model assumes that item effects are the same for all subjects of a benchmark. With no labels, it predicts close to the average level of a group, since the accuracy of a subject on a new benchmark is hard to predict from the data we have. The Qwen parts need a GPU at test time; without one, the predictor still runs with the e5 model alone.

What did not help: We tested these ideas with the same paired simulation and dropped them.

- Predicting budget 0 from item text and subject details (provider, release date, size, model name) with LightGBM [Ke et al., 2017]. The best B_0 was 0.248, against 0.250, while the centred prior reaches 0.2356.
- Asking Qwen2.5-7B to rate item difficulty from 1 to 9. The rank correlation with measured difficulty inside a benchmark was -0.03.
- Larger item vectors (e5-base, e5-large, other Qwen layers). None beat e5-small: with so few labels, the number of labels limits the model more than the quality of the vectors.
- A prior on subject ability from the same model on other benchmarks. Its out-of-fold correlation with ability on held-out benchmarks was -0.16, and it gave no gain.
- Learned calibration (stacking with LightGBM, Platt scaling [Platt, 2000] per budget). Both were worse on held-out benchmarks, because calibration learned on some benchmarks does not carry over to others.
- Choosing labels by uncertainty, overlap with other subjects, embedding diversity or D-optimal design. None was clearly different from random order.

7 Related Work

Item response theory: IRT [Lord, 2012] models a correct answer from the ability of a subject and the difficulty of an item. tinyBenchmarks [Maia Polo et al., 2024] uses IRT to estimate the accuracy of a language model from a small set of items, with item parameters learned from models already run on the same benchmark. GP-IRT instead learns the item effects of a new benchmark at test time, and links items through a Gaussian process prior on their embeddings.

Gaussian process classification: GP-IRT uses the Laplace approximation for Gaussian process classification [Rasmussen and Williams, 2006] and the probit approximation of the predictive integral [MacKay, 1992]. Averaging predictions over prior scales with weights from the marginal likelihood follows Bayesian model averaging [Hoeting et al., 1999].

8 Conclusion

In this paper, we designed GP-IRT, a method that predicts whether an AI system answers an item of a new benchmark correctly from a few labels. GP-IRT fits a logistic item response model on each benchmark at test time, uses the labels of all subjects together, links items through a Gaussian process prior on their embeddings, and averages over prior scales. In a simulator of the official protocol on held-out benchmarks, GP-IRT reaches a Brier ALC of 0.1797, and centring the prior on the benchmark level gives the largest single gain. Most of the remaining error is at the level of the group, which is hard to predict for a new benchmark from the data we have.

Reproducibility: The code is at `https://github.com/nguefackuriel/paiec-gpirt`. Its README gives the steps to rerun everything: download and preparation of the training pool, embeddings, the simulator, the scripts behind each table and figure, and the build of the submission. Pretrained models: `intfloat/multilingual-e5-small` and `Qwen/Qwen2.5-7B-Instruct`. Training data: only `aims-foundations/measurement-db`.

Use of AI assistants: The code, the experiments and drafts of this report were made with the help of an LLM coding assistant (Claude Code, Anthropic). All results come from the code in the repository.

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